

Linking the Integrated Management of Childhood Illness (IMCI) and health information system (HIS) classifications: issues and options

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Differences in the terms used to classify diseases in the Integrated Management of Childhood Illness (IMCI) guidelines and for health information system (HIS) disease surveillance could easily create confusion among health care workers. If the equivalent terms in the two classifications are not clear to health workers who are following the guidelines, they may have problems in performing the dual activities of case management and disease surveillance. These difficulties could adversely affect an individual's performance as well as the overall effectiveness of the IMCI strategy or HIS surveillance, or both. We interviewed key informants to determine the effect of these differences between the IMCI and HIS classifications on the countries that were implementing the IMCI guidelines. Four general approaches for addressing the problem were identified: translating the IMCI classifications into HIS classifications; changing the HIS list to include the IMCI classifications; using both the IMCI and HIS classification systems at the time of consultations; and doing nothing. No single approach can satisfy the needs of all countries. However, if the short-term or medium-term goal of IMCI planners is to find a solution that will reduce the problem for health workers and is also easy to implement, the approach most likely to succeed is translation of IMCI classifications into HIS classifications. Where feasible, a modification of the health information system to include the IMCI classifications may also be considered.

Keywords: child health services; delivery of integrated health care; disease classification; information systems; translation guidelines.

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Background

The WHO/UNICEF strategy for the Integrated Management of Childhood Illness (IMCI) is now being used by health care workers in a number of countries in patients' homes and in health facilities. The aim of the strategy is to reduce childhood deaths, illnesses and disabilities and to contribute to the improved growth and development of children in developing countries. An important component of the IMCI strategy is the application of standard clinical guidelines in the management of sick children who are brought to a health facility. The current version of the IMCI guidelines, which covers children aged from 1 week to 5 years, is based on the detection of simple clinical signs without resorting to any laboratory tests, followed by empirical treatment. Using these guidelines, the

health care worker assesses ill children in a systematic way and then classifies them as to whether they require urgent referral and immediate pre-referral treatment, or specific medical treatment and advice, or simple advice and home management. The IMCI strategy includes other interventions relating to the health system and seeks to improve family and community practices. As of June 1999, a total of 63 countries had started implementing the IMCI strategy; 12 of them were already in an expansion phase and were facing problems caused by incompatibility between their health information system (HIS) for disease surveillance and the IMCI disease classifications.

When IMCI guidelines are applied in a country, common diagnoses of childhood illness, such as bronchitis and meningitis, are replaced by terms given in the IMCI classifications, and a problem can arise when the HIS terms for disease surveillance do not match with those in the IMCI classifications. For example, a health care worker dealing with a child who has fever and neck stiffness may suspect meningitis, which is a reportable disease in most countries; however, IMCI classifies this illness as "very severe febrile disease" (a term not likely to appear on an HIS list of reportable diseases). In this instance, the IMCI classification is not compatible with the HIS diagnosis. As a result, health workers may have difficulty in deciding whether to report the

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illness as "meningitis" or as "very severe febrile disease". This type of confusion could easily affect a health worker's performance.

Before their implementation in a country, the generic IMCI guidelines have to be adapted to local conditions by a national IMCI working group. If any IMCI-HIS incompatibilities are suspected, the working group can use the adaptation process to eliminate potential problems before they arise. Although WHO has produced a guide to help with the adaptation process, the current version of this guide does not include specific advice on how to address IMCI-HIS incompatibility (1). The present article describes some problems due to these differences and evaluates the options for addressing them.

Methods

Data collection and definitions

A total of 14 key informants from WHO, Geneva, Switzerland, and the Centers for Disease Control and Prevention, Atlanta, GA, USA, were interviewed between December 1997 and January 1998 about their experiences in applying IMCI in nine countries (Egypt, Ethiopia, Kenya, Madagascar, Morocco, Peru, Uganda, United Republic of Tanzania, and Viet Nam), as well as about their general field experience in developing countries. Unpublished reports documenting IMCI implementation in countries were also used.

For this study, a *classification* was defined as a functional label that describes a constellation of symptoms, signs, laboratory findings, or results of imaging studies. A *diagnosis* was defined as a nosological term that describes a disease affecting an individual. For example, in a person with symptomatic *Vibrio cholerae* infection, the term "diarrhoea" could be used to describe having three or more liquid bowel movements in a 24-hour period, while "cholera" is the term for the diagnosis. According to the IMCI guidelines, *danger signs* are defined as the inability to drink, vomiting everything, the presence of convulsions, and lethargy or unconsciousness (2).

A *routine* disease reporting system is defined as a surveillance system that collects data at regular intervals, typically every 1–3 months. An *immediate* reporting system is defined as a surveillance system that collects data on a disease within a few days of identification. Immediate reporting systems are often informal, e.g. when a health worker reports a case of cholera by telephone to the district health officer.

Assumptions

Because disease surveillance systems vary greatly among countries, the following generic disease surveillance system is used to facilitate discussion. During consultations, the health worker records the diagnoses or classifications in one or more patient registers. On a regular basis, data from the patient

registers are transferred to an HIS routine summary reporting form. The forms from individual health facilities are sent to a higher administrative level every month. It is assumed that a formal or informal immediate reporting system exists for diseases that have important public health implications and that the health information system uses disease surveillance classifications that are typical of surveillance systems in developing countries (Table 1). The case definitions for these classifications are based upon WHO recommendations (3). With the exception of tuberculosis, all case definitions on this generic list can be satisfied by clinical criteria alone, and are therefore appropriate for use in outpatient health

Table 1. Comparison of disease classifications using the Integrated Management of Childhood Illness (IMCI) guidelines and a typical disease surveillance system

General disease category	IMCI classification	Disease surveillance classification ^a
Respiratory diseases	Severe pneumonia Pneumonia No pneumonia: cough or cold	Diphtheria Pertussis Pneumonia Tuberculosis
Diarrhoeal diseases	Diarrhoea with severe dehydration Diarrhoea with some dehydration Diarrhoea with no dehydration Severe persistent diarrhoea Persistent diarrhoea Dysentery	Bacillary dysentery Cholera Diarrhoea without blood
Diseases with fever	Very severe febrile disease Malaria Fever: malaria unlikely Severe complicated measles Measles with eye or mouth complications Measles	Dengue fever Malaria Measles Meningitis Plague Yellow fever
Ear infections	Mastoiditis Acute ear infection Chronic ear infection No ear infection	Ear infection
Nutritional disorders	Severe malnutrition or severe anaemia Anaemia or very low weight No anaemia and not very low weight	Anaemia Malnutrition
Other conditions		Dental diseases Dracunculiasis Neonatal tetanus Other disease ^b Poliomyelitis Trauma

^a Classifications are listed alphabetically for each general disease category; no correspondence is implied between the disease surveillance classification and the IMCI classification shown on the same row of the table.

^b Classification for any diseases not already classified.

facilities where there are few diagnostic instruments and no laboratory facilities. Furthermore, it is assumed that IMCI-trained health workers are able to classify children using the IMCI guidelines with a high degree of accuracy.

Incompatibility between IMCI and HIS classifications

Cause of incompatibility

The terms used in the IMCI and HIS classifications differ because the two systems have different objectives. The primary aim of the IMCI strategy is to prevent deaths and disabilities by improving the case management of sick children in outpatient health facilities. The IMCI guidelines have therefore been focused on the treatment of individual children. In contrast, the purpose of the HIS classification is to provide disease surveillance data to guide public health policy, and it therefore focuses on the health of populations.

Because of their different objectives, the IMCI and HIS classifications were designed in different ways. IMCI classifications were designed to guide health care workers to follow a treatment plan for an ill child presenting with a specific set of symptoms and signs. The IMCI design has four key features.

- The classifications were aimed for use in a setting that had few diagnostic tools, and no laboratory or imaging facilities.
- The classifications needed to have a relatively high sensitivity for diagnoses that cause the greatest proportion of childhood mortality.
- Etiologically distinct diagnoses were combined when the treatment was the same. For example, the "pneumonia" classification included pneumonia caused by either *Streptococcus pneumoniae* or *Haemophilus influenzae* because both can be treated effectively with a 5-day course of oral antibiotics.
- Separate classifications were given for the same diagnosis when the disease had different levels of severity. For example, pneumococcal pneumonia that causes only cough and tachypnoea was classified as "pneumonia", while pneumococcal pneumonia that causes cough and stridor was classified as "severe pneumonia or very severe disease".

An important consequence of this design is that some IMCI classifications are poor predictors of the underlying diagnosis. For example, a child with the classification "very severe febrile disease" could have meningitis (viral or bacterial), or sepsis (from a number of possible pathogens), or cerebral malaria.

In contrast, the classifications (or case definitions) used by health information systems were designed to detect etiologically distinct diseases. For example, the immediate reporting component of a health information system in West Africa would probably be designed to detect cholera, not simply diarrhoea. Because of the public health (and even

political and economic) implications of detecting certain diseases, some HIS classifications are so specific that they actually establish the diagnosis. For example, the WHO-recommended case definition for sporadic cases of Ebola haemorrhagic fever includes confirmation with the same "gold standard" laboratory tests that are used to establish the diagnosis in a hospitalized patient (3). Because the designers of health information systems recognized that diagnoses could rarely be made by health workers in outpatient health facilities, the HIS classifications for these settings may be simplified. These simplified classifications are useful for detecting suspected cases of diseases under surveillance (3, 4). These differences in the design of the IMCI and HIS classifications led to the incompatibilities between them.

Specific incompatibilities between the IMCI and HIS classifications

Some areas of incompatibility between the two classifications are described below.

- The IMCI and HIS classifications often use different names, although in some instances the differences are minor, such as "severe pneumonia" (IMCI) and "pneumonia" (HIS). Only four classifications are the same on both lists: anaemia, malaria, measles, and pneumonia (Table 1).
- Some IMCI classifications satisfy the case definitions for two or more HIS classifications (Table 2). For example, the IMCI classification "very severe febrile disease", which is applied to a child with fever and either neck stiffness or a danger sign, satisfies the case definitions for both "meningitis" and "malaria" in the HIS classification.

This type of incompatibility can be particularly troubling to health workers and persons responsible for HIS management and IMCI management. Health workers may be confused by having to choose between two HIS diagnoses that are quite different (e.g. meningitis and malaria). The HIS management may be concerned because the positive predictive value of the surveillance system will decrease for some diseases. For example, in a country with endemic malaria, all children presenting with fever will be given the IMCI classification "malaria", even if the true cause of the fever is malaria, a respiratory infection, or dysentery. Thus, there may be difficulties in interpreting disease trends for malaria after IMCI implementation. These difficulties may be particularly troubling during the meningococcal meningitis season, when epidemiologists use surveillance data to determine whether an epidemic has started and whether a vaccination campaign should be initiated.

The diagnostic accuracy of health workers in outpatient facilities is, however, probably low for diseases with nonspecific symptoms. Therefore, in practical terms, the impact of this type of IMCI-HIS incompatibility on the quality of routine disease surveillance data is limited.

- The following IMCI terms have no corresponding HIS classification: “no pneumonia: cough or cold”, “fever — malaria unlikely”, “mastoiditis”, “no ear infection”, and “no anaemia and not very low weight”. This type of incompatibility does not pose a serious problem because any term describing the absence of a disease, such as “no ear infection”, need not be reported at all; other terms in the IMCI classification could be reported under the HIS classification “other diseases”.
- The following HIS diagnoses have no corresponding IMCI classification: “dental diseases”, “dracunculiasis”, “neonatal tetanus”, “poliomyelitis”, and “trauma”. The reason for this difference is that the IMCI classification and treatment guidelines only relate to children who present with a danger sign or a “main symptom” (cough or difficult breathing, diarrhoea, or fever). For conditions that do not cause a danger sign or a main symptom, such as trauma, health workers should proceed directly to an HIS classification.
- Some HIS diagnoses, such as tuberculosis, require laboratory confirmation. Because the case definition cannot be satisfied at most outpatient facilities, tuberculosis does not represent an incompatibility between the two classification systems. This argument applies to any HIS classification that requires a laboratory or radiographic result to satisfy the case definition.
- There are IMCI–HIS incompatibilities that are not a direct result of the different disease classification systems. These differences, however, can still confuse health workers who have to perform the dual activities of case management and disease surveillance. For example, the way in which a child’s age is reported may differ between the two systems. According to IMCI guidelines, if the infant is <2 months old, the age is recorded in weeks; if the child is ≥2 months, the age is recorded in months. Among HIS classifications, there are a variety of different systems for reporting age, such as using months if it is <12 months and years if it is ≥12 months. IMCI and HIS may also have different policies regarding the following: whether diseases should be reported among children who return to a health facility for a follow-up visit; whether diseases should be reported among children who are referred to a hospital; and the number of diseases that can be reported for each child.

Addressing IMCI–HIS incompatibility

Several options exist for resolving IMCI–HIS incompatibilities, including the following: 1) translating IMCI classifications into HIS classifications; 2) changing the HIS list to include IMCI classifications; 3) using both IMCI and HIS classification systems at the time of consultation; and 4) doing nothing.

Table 2. Integrated Management of Childhood Illness (IMCI)–health information system (HIS) incompatibilities (the symptoms and signs leading to one IMCI classification satisfy the case definitions for two or more diagnoses by the HIS classification)

The child’s only symptoms and signs	IMCI classification	HIS classification ^a
1. Cough or difficult breathing	No pneumonia: cough or cold	Diphtheria, or pertussis
2. Cough and stridor	Severe pneumonia	Pneumonia, or pertussis
3. Difficult breathing and chest wall indrawing	Severe pneumonia	Pneumonia, or diphtheria
4. Difficult breathing and lethargy or unconsciousness	Severe pneumonia ^b	Pneumonia, or plague
5. Acute watery diarrhoea	Diarrhoea with no, some, or severe dehydration	Diarrhoea without blood, or cholera ^c
6. Fever and a stiff neck	Very severe febrile disease	Meningitis, or malaria
7. Fever and unconsciousness	Very severe febrile disease	Meningitis, or malaria, or plague
8. Fever, difficult breathing and unconsciousness	Very severe febrile disease and severe pneumonia ^b	Pneumonia, or pertussis, or plague, or malaria, or meningitis
9. Severe visible wasting and severe palmar pallor, or very low weight and non-severe palmar pallor	Severe malnutrition and severe anaemia, or very low weight and anaemia	Malnutrition, or anaemia

^a Case definitions based on WHO recommendations (3).

^b Just before death, most diseases cause difficult breathing and a decreased level of consciousness (the IMCI classification for all such scenarios would be “severe pneumonia”).

^c Assuming the cholera classification can be applied to children <5 years of age.

With the first of the above-mentioned options, the health worker classifies children using the IMCI classification at the time of the consultation and records this classification in a patient register. At the end of each day, the health worker (or other staff member) translates the IMCI classifications into HIS classifications using a standard translation table (Table 3), and the HIS classifications are reported. The primary advantage of this option is that it simplifies the task of the health worker at the time of the consultation. A second advantage is that this option can be implemented without modifying the way in which surveillance data are collected by the health information system. The disadvantages of this option are that translation is additional work; special training is needed for those performing the translation; errors in translation may occur; and, depending on how the translation table is designed, certain HIS classifications may not appear in it at all, and first-level health facilities may cease reporting these HIS classifications for children under 5 years of age. In addition, because IMCI classifications often combine distinct diagnoses, translating IMCI classifications into HIS classifications may significantly change the way in which surveillance data are interpreted and used. For example, if the IMCI classification “very severe febrile disease” were translated into the HIS classification

Table 3. Some examples for translating Integrated Management of Childhood Illness (IMCI) classifications into health information system (HIS) classifications

IMCI classification	HIS classification
Severe pneumonia or very severe disease	Pneumonia
Pneumonia	Pneumonia
No pneumonia: cough or cold	Other diseases
Diarrhoea with severe dehydration	Diarrhoea without blood
Diarrhoea with some dehydration	Diarrhoea without blood
Diarrhoea with no dehydration	Diarrhoea without blood
Severe persistent diarrhoea	Diarrhoea without blood
Persistent diarrhoea	Diarrhoea without blood
Dysentery	Bacillary dysentery
Very severe febrile disease	Malaria or meningitis ^a
Malaria	Malaria
Fever — malaria unlikely	Other diseases
Severe complicated measles	Measles
Measles with eye or mouth complications	Measles
Measles	Measles
Mastoiditis	Ear infection
Acute ear infection	Ear infection
Chronic ear infection	Ear infection
No ear infection	Not reported
Severe malnutrition or severe anaemia	Malnutrition or anaemia ^b
Anaemia or very low weight	Malnutrition or anaemia ^b
No anaemia and not very low weight	Not reported

^a In a specific country, only one of these would be included on the translation table.

^b Both classifications are reported if the child had both malnutrition and anaemia. Otherwise the health care worker reports only the individual classification that is applicable (malnutrition or anaemia).

“meningitis”, it would be difficult to determine whether an increase in the number of reported cases of meningitis were caused by an epidemic of dengue fever, malaria, meningitis, or plague.

A variation of the first option is to carry out a computer translation at the district or national level. The advantages are that health workers do not need to perform the translation, the need for special translation training is eliminated, and the quality of the translation will not be affected by human error. The disadvantages are that the IMCI classifications must be reported to the district or national level — essentially this calls for an expansion of the existing reporting system — and that the process of reporting IMCI classifications (i.e. copying information from patient registers to reporting forms) and data entry may introduce errors in the translations.

With the second of these options, the health information system is modified to accommodate IMCI classifications. Illnesses not addressed by IMCI, such as trauma or acute flaccid paralysis, are reported according to the pre-existing HIS policy.

The advantages of this option are that all IMCI–HIS incompatibilities are eliminated, and data on reported IMCI classifications would be available to IMCI programme managers via the health information system. These data may be useful for programme planning (such as estimating drug requirements) and monitoring and improving the quality of case management. Including IMCI classifications in the HIS list might also increase the visibility and importance of IMCI and reinforce correct application of the guidelines. Furthermore, for the purposes of disease surveillance, IMCI classifications by syndromes have two advantages: they may be more reliable than diagnoses by health workers because the IMCI classification process involves less clinical judgement; and IMCI classifications may provide more specificity regarding the severity of the illness.

Option two also has important disadvantages. First, the addition of a dozen or so IMCI classifications will increase the complexity of the health information system, and increased complexity may lead to dysfunction. Second, as with option one, surveillance of IMCI classifications, which are often poor predictors of specific diagnoses, may significantly change the way in which surveillance data are interpreted and used. Third, a concern associated with the implementation of option two is that if the health information system is changed before most health workers in a country have been trained in IMCI, the system may face further difficulties in interpreting diseases reported by untrained staff. If untrained health workers reported HIS classifications, the health information system would have a mix of HIS and IMCI classifications, and if untrained workers attempted to use IMCI classifications, most would be incorrect. A final concern is that HIS managers may be reluctant to make substantial changes in the way data are collected and interpreted, and without their active support it is impossible to implement this option.

With the third of the above options, the health worker classifies children using both systems at the time of the consultation; the IMCI classification is used to guide treatment, and HIS classifications are used for reporting. The advantages of this option are that all IMCI–HIS incompatibilities are eliminated, and no modification of the health information system is needed. The major disadvantages are that health workers may be confused by the simultaneous use of two classification systems, and some children would need to be classified twice. The increased effort and time required to perform dual evaluations may undermine the continued correct use of IMCI and HIS classifications.

A variant of option three calls for the health worker to classify children using both systems at the time of the consultation and to report both sets of classifications in parallel reporting systems. This option is equivalent to the policy common to many vertical disease control programmes of adding a separate programme-specific surveillance system for the disease addressed by the programme. The

advantages of this approach are essentially the same as those listed above for options two and three; namely, all incompatibilities are eliminated, and data on IMCI classifications are available to IMCI managers for monitoring and planning purposes. Similarly, the disadvantages of this approach are common to options two and three, including an increased complexity and workload for health workers and a greater time required per child.

With the fourth of the above options, no change is made and no standard advice is given to health workers on how to cope with IMCI–HIS incompatibilities. If no effort is made to address these incompatibilities during IMCI planning, this option clearly becomes the default. The major disadvantage with this approach is that health workers may become confused and frustrated, and may either stop using the IMCI classification or stop reporting to the health information system. Furthermore, experience in countries that have already begun implementing IMCI has revealed that having no policy to address the incompatibility results in a heterogeneity of methods created by the health workers, thus making HIS data difficult to interpret. For example, in one country, some health facility clerks, who had the responsibility for transferring patient register data to a monthly HIS reporting form, recorded all incompatible diagnoses (such as “very severe febrile disease”) under the “other diseases” category. As a result, the incidence of cases of certain reportable diseases, such as meningitis, dropped precipitously after IMCI was implemented.

Discussion

The different objectives of IMCI and HIS have led to different designs for their disease classifications; as a result, the two systems are not always compatible. Without some clarification or an “interface” linking the two systems, health workers using IMCI may have difficulties in performing the dual activities of case management and disease surveillance. These difficulties could adversely affect an individual health worker’s performance as well as the overall effectiveness of IMCI or HIS, or both.

The planning phase of IMCI implementation provides the opportunity to anticipate and resolve IMCI–HIS incompatibilities before they cause difficulties for health workers. The variations in health infrastructures, IMCI adaptations, and health information systems are so great worldwide that no single solution will satisfy the needs of all countries. However, if the short-term or medium-term goal of IMCI planners is to choose an approach that reduces the confusion among health workers and is easy for IMCI programme managers to implement, translating IMCI classifications into HIS classifications (option one mentioned above) is the approach most likely to succeed. This option is particularly appropriate during the early stages of IMCI, while managers and health workers are gaining experience

in applying the IMCI classifications, and before all health workers have been trained. In countries where HIS managers are willing to make modifications, adding IMCI classifications to the HIS list (option two) may be considered.

The primary justification for either option one or two is that the difficulties caused by IMCI–HIS incompatibility will not affect health workers during consultations with children, and therefore should not affect the quality of case management. Advantages of option one are that the health information system does not need to change its classification system and that other incompatibilities (such as how a child’s age is reported and whether the illnesses of referred children and children coming for a follow-up visit should be reported) can also be resolved at the time of translation. In addition, even if the health information system can be modified, the translation option can be used as a temporary solution while IMCI managers are gaining experience and HIS managers are planning how best to implement the changes.

The other approaches for addressing IMCI–HIS incompatibility described in this article have major disadvantages. As already mentioned, the problem with modifying the HIS list to include IMCI classifications (option two) is that it may be unrealistic to expect the health information system to make such far-reaching changes. However, it should be noted that the Integrated Disease Surveillance initiative in Africa has been striving to make the health information system more flexible and responsive to programmatic needs (4). To date, Morocco is the only country to use this approach, and it will be important to follow progress to see how the new health information system is functioning and whether surveillance of IMCI classifications is useful in guiding public health policy.

The problem with using both the IMCI and HIS classifications at the time of a consultation (option three) is that it may increase, rather than decrease, the workload and confusion among health workers during consultations with ill children. The problem with doing nothing (option four) is that the confusion among health workers during consultations will remain, and this may lead them to resolve the IMCI–HIS incompatibilities in a variety of unpredictable ways. These differences in the use of HIS classifications may make it impossible to interpret disease surveillance data.

There are some disadvantages in the translation approach. A potentially troubling problem is that reporting from first-level health facilities might cease for certain HIS classifications for children under 5 years of age (this is also a problem for option two). For example, in Table 3, diphtheria, pertussis, tuberculosis, and plague have not been included. These exclusions were made for two reasons: each IMCI classification can only be translated to one HIS classification; and the IMCI assessment and classification system does not provide enough information to distinguish these HIS classifications from the

other, more general, respiratory classifications that are included in Table 3. The loss of routine reporting of these classifications for children under 5 years of age from first-level health facilities can be justified because some severely ill children will be taken to a hospital where the disease will be diagnosed and reported; the ability of the health information system to detect epidemics, except for neonatal tetanus, would not be compromised because diseases with epidemic potential almost always affect older children and adults, on whom disease surveillance would continue; and health workers can always use a formal or informal immediate disease reporting system for reporting suspected cases of diseases with special public health importance.

Other drawbacks of the translation approach include the requirement for additional training of health workers in using the translation, and the actual work involved in translating IMCI classifications to HIS classifications. From a global perspective, an additional disadvantage of option one is that if different countries develop different translation

guidelines, the task of interpreting disease surveillance data becomes more complicated.

In conclusion, the process of IMCI implementation is complex, and there are many obstacles to be overcome. If IMCI planners could anticipate the incompatibilities between their programme and the country's health information system, a relatively small investment in time and effort spent in analysing the problem and in communicating with the HIS management could help reduce the confusion among health workers who will be performing disease surveillance duties while applying the IMCI guidelines in children's clinics. ■

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Résumé

Relier les classifications établies aux fins de la prise en charge intégrée des maladies de l'enfant (PCIME) et des systèmes d'information sanitaire (SIS) : problèmes et solutions

La stratégie OMS/UNICEF de prise en charge intégrée des maladies de l'enfant (PCIME) est très largement utilisée par les agents de santé au domicile des malades et dans les centres de santé. Son but est de réduire les taux de morbidité, d'incapacités et de mortalité chez les enfants et de contribuer à améliorer la croissance et le développement des enfants dans les pays en développement. L'un des éléments importants de cette stratégie consiste à appliquer des recommandations cliniques générales à la prise en charge des enfants malades amenés à un centre de santé.

Les objectifs de la PCIME et de la composante surveillance des maladies d'un système d'information sanitaire (SIS) sont différents. D'une manière générale, la PCIME vise à améliorer les soins donnés à des enfants malades pris individuellement tandis que les SIS sont conçus pour détecter des cas de maladies déterminées. A cause de ces différences, les classifications des maladies correspondant à ces deux axes d'activités ne sont pas toujours compatibles. En l'absence, entre ces deux classifications, de lien clairement défini ou d'interface permettant de mieux comprendre des termes équivalents, les agents de santé qui appliquent les recommandations de la PCIME risquent d'éprouver des difficultés à remplir leur double fonction de prise en charge des cas et de surveillance des maladies. De telles difficultés pourraient compromettre leur travail ainsi que l'efficacité globale de la stratégie PCIME, ou les activités de surveillance des maladies des SIS, ou encore les deux.

Nous avons interrogé des spécialistes de la question à l'OMS et aux Centers for Disease Control and Prevention des Etats-Unis d'Amérique afin de déterminer comment les pays qui mettent en œuvre la

PCIME ont pu être gênés par d'éventuelles incompatibilités entre les classifications PCIME et SIS. A partir des expériences de ces pays et d'hypothèses, quatre solutions ont été proposées : convertir les classifications PCIME en classifications SIS; changer les systèmes d'information sanitaire afin d'y inclure les classifications PCIME; utiliser à la fois la classification PCIME et la classification SIS au moment de la consultation; enfin, laisser les choses en l'état. Les avantages et les inconvénients de chacune de ces solutions sont analysés en détail dans l'article.

Dans le monde, les variations entre les systèmes d'information sanitaire, les adaptations de la PCIME et les infrastructures de santé sont telles qu'aucune solution unique ne saurait répondre aux besoins de tous les pays. Toutefois, si, à court ou à moyen terme, les responsables de la PCIME souhaitent une formule qui atténue les difficultés des agents de santé et soit facile à appliquer, la solution offrant les meilleures garanties de succès serait de convertir les classifications PCIME en classifications SIS. Elle serait particulièrement appropriée aux premiers stades de la PCIME, à un moment où les administrateurs et les agents de santé se familiarisent avec la classification PCIME et où tous les agents de santé n'ont pas encore été formés. Les pays qui peuvent modifier leur système d'information sanitaire pourraient envisager d'ajouter la classification PCIME à la liste SIS. En consultation avec la direction du SIS, les responsables de la PCIME pourraient aussi concevoir des moyens de lever les ambiguïtés qui gênent les agents de santé chargés d'exécuter des activités de surveillance tout en appliquant les recommandations de la PCIME sur le terrain.

Resumen

Vinculación entre la Atención Integrada a las Enfermedades Prevalentes de la Infancia (AIEPI) y las clasificaciones del sistema de información sanitaria (SIS): problemas y opciones

La estrategia OMS/UNICEF de Atención Integrada a las Enfermedades Prevalentes de la Infancia (AIEPI) es ampliamente utilizada por los agentes de salud en los hogares de los pacientes y los centros sanitarios. El objetivo de la estrategia es reducir las enfermedades, las discapacidades y las defunciones infantiles y contribuir a mejorar el crecimiento y desarrollo de los niños en los países en desarrollo. Un componente importante de la estrategia de AIEPI es la aplicación de directrices clínicas estándar al manejo de los niños enfermos que llegan a los establecimientos de salud.

Los objetivos de la estrategia AIEPI y del componente de vigilancia de enfermedades de los sistemas de información sanitaria (SIS) son diferentes. En general, la AIEPI aspira a mejorar el tratamiento de los niños enfermos, mientras que los SIS se han diseñado para detectar la aparición de enfermedades específicas. Debido a esos diferentes objetivos, las dos clasificaciones de enfermedades resultantes no siempre son compatibles. Si no hay ningún vínculo o punto de contacto claro entre las dos clasificaciones para facilitar la identificación de los términos equivalentes, los agentes de salud que utilizan las directrices de AIEPI pueden encontrar problemas en su doble actividad de manejo de casos y vigilancia de las enfermedades. Estas dificultades pueden perjudicar el desempeño individual y la eficacia global de la estrategia de AIEPI o la vigilancia de las enfermedades por el sistema de información sanitaria, si no ambas cosas.

Entrevistamos a las personas pertinentes en la OMS y en los Centros de Control y Prevención de Enfermedades de los Estados Unidos (CDC) para determinar en qué medida se habían visto afectados por las incompatibilidades AIEPI-SIS los países que

estaban aplicando la AIEPI. A partir de las experiencias de estos países y de consideraciones teóricas se formularon cuatro enfoques generales para abordar el problema: la traducción de las clasificaciones de la AIEPI en las clasificaciones del SIS; la modificación del sistema de información sanitaria para incluir en él las clasificaciones de la AIEPI; el uso de las dos clasificaciones, AIEPI y SIS, en el momento de la consulta; y la alternativa de no tomar ninguna medida. En el artículo se analizan las ventajas e inconvenientes de cada opción.

Las diferencias entre los sistemas de información sanitaria, las adaptaciones de la AIEPI y las infraestructuras de salud son tan importantes en todo el mundo que no es posible hallar una solución que satisfaga las necesidades de todos los países. Sin embargo, si la meta a corto o medio plazo de los planificadores de la AIEPI estriba en aplicar un enfoque que ahorre problemas a los agentes de salud y sea de fácil aplicación, el enfoque más prometedor es la traducción de la clasificación de la AIEPI en las clasificaciones del SIS. Esta opción es especialmente apropiada para las fases iniciales de la AIEPI, mientras los gerentes y el personal sanitario adquieren experiencia en el uso de la clasificación de la AIEPI, y antes de que se haya capacitado a todos los profesionales sanitarios. En los países que pueden modificar su sistema de información sanitaria, puede considerarse la posibilidad de añadir las clasificaciones de la AIEPI a la lista del SIS. Los planificadores de la AIEPI, trabajando en consulta con los administradores del SIS, podrían también elaborar una política que eliminase la confusión entre los agentes de salud que han de desempeñar funciones de vigilancia mientras aplican las directrices de la AIEPI sobre el terreno.

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